ADITYA GAUTAM

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SUMMARY

Electronics and VLSI Engineering student at IIIT Delhi with experience in embedded AI, FPGA design, and Python simulation. Proficient in C++, Python, and Verilog; interested in software development, AI, and intelligent systems.

EDUCATION

B.Tech, Electronics and VLSI Engineering, IIIT Delhi — 2024 - 2028
CGPA: 7.19 (till 4th Semester)

CBSE Class 12, PCM, Green Fields School, New Delhi — 2022 - 2023
Percentage: 89%

CBSE 10th Board, Green Fields School, New Delhi — 2020 - 2021
Percentage: 96%

SKILLS

Technical Skills	DSA, Microcontrollers, Digital Logic, PCB Design, FPGA, HW-SW Co-design, Xilinx Vivado, Vitis HLS, KiCad, LTspice, Arduino IDE, STM32CubeIDE, Figma (UI/UX)
Programming Languages	C++, Python, C (Embedded), Verilog
Soft Skills	Problem-solving, Team collaboration, Technical communication, Research, Adaptability

EXPERIENCE

Internship Trainee — Jun 2025 - Jul 2025
ISSA DRDO — Team Size: 2

- Developed a Python-based mathematical model for Electronic Support Measures (ESM), simulating electronic warfare threat parameters including PRI, PW, RF, and AOA.

Sales and Marketing Intern — April 2021
Homeflic Wegrow

- Executed sales and marketing campaigns; strengthened lead generation and market research skills.

PROJECTS

RISC-V 32I Assembler & Simulator | *Python, RISC-V, Bit Manipulation* — Feb - Apr 2025
Team Size: 4

- Built a RISC-V 32I assembler (instruction parsing, opcode generation, label resolution) and an instruction-level simulator (registers, memory, control flow); verified supported instructions using systematic test cases. Assembler Simulator

GRAIL: Gesture-Controlled Robotic Hand with On-Device AI Inference — Jul - Aug 2026
Python, C (Embedded), STM32CubeIDE, X-CUBE-AI, TensorFlow, MediaPipe, KiCad
Team Size: 4 | GitHub

- Designed and engineered a vision-based gesture-controlled robotic hand, executing full on-device neural-network inference on an STM32H7 microcontroller.
- Trained a custom dense neural network on 21-point MediaPipe hand-landmark data and deployed via X-CUBE-AI with INT8 quantization, fitting within 3 KB of RAM at 99.3% classification accuracy.
- Migrated architecture from ESP32 to STM32H7 to enable edge inference, resolving low-level firmware, pin-mapping, and data-quantization constraints.

Mathematical Model of ESM | *Python, NumPy, Matplotlib* — Jun - Jul 2025

- Built modular emitter, parameter-extraction, threat-library, and visualization components; displayed animated polar tracks with emitter IDs and RF/PRI/PW/AOA readouts. GitHub

Embedded Logic Design | *FPGA, Vitis HLS, PYNQ, HW/SW Co-design* — Oct - Dec 2025

- Designed matrix multiplication IP using HLS and tested on FPGA hardware; implemented radar range estimation and CNN inference via HW-SW co-design on PYNQ. Link1 Link2

VOLUNTEER EXPERIENCE

- **Events Team, Esya:** Organised the Line Follower event at IIITD Tech Fest ESYA. — Jul - Aug 2025
- **Mentor, Metamorphosis 2026:** Mentored students in an innovation program by The Tale of Humankind Foundation in collaboration with IIT Delhi Office of Academic Outreach. — May 2026